

Briana Aniece Sealey, Ph.D.

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Education

- Ph.D. Ecology, Evolution and Behavior, The University of Texas at Austin (UT Austin)** 2019 – 2024
Dissertation: Evaluating antipredator behavior and food preferences in the Jamaican fruit-eating bat, *Artibeus jamaicensis*
- M.S. Ecology and Evolutionary Biology, University of Michigan, Ann Arbor (UM)** 2017 – 2019
Thesis: Predator cues and ontogeny drive variation in anti-predator displays of South American Calico snakes (*Oxyrhopus*)
- B.A. Biology, Case Western Reserve University (CWRU)** 2013 – 2017
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Academic Appointments

- Graduate Teaching Assistant, The University of Texas at Austin** 2023 – 2024
Department of Biology
- Graduate Research Fellow, Smithsonian Tropical Research Institute (STRI)** 2019 – 2024
Gamboa Laboratories and Barro Colorado Island
- Adjunct Instructor, Huston-Tillotson University** Fall 2020
Department of Biology
- Graduate Student Researcher, University of Michigan, Ann Arbor** 2017 – 2019
Department of Ecology and Evolutionary Biology
- Graduate Student Instructor, University of Michigan, Ann Arbor** 2017 – 2019
Department of Biology
- Undergraduate Research Assistant, Case Western Reserve University** 2016 – 2017
Department of Biology
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Teaching Experience

- Graduate Teaching Assistant, The University of Texas at Austin** 2023 – 2024
Molecular Biology (BIO 311C), 40 students
Ecology, Evolution & Behavior (BIO 311D), 80 students
Department of Biology
Supported instructor by leading discussion sections that synthesized learning material from lecture and evaluated student learning by grading exams and written reports.
- Adjunct Instructor, Huston-Tillotson University** Fall 2020
Ecology and Behavior (BIO 2430), 18 students
Department of Biology
Instructor of record for introductory course, developed 50-minute lectures three times per week, and 90-minute online lab experiments one time per week. Designed laboratory experiments to integrated learning material from lecture and

assessed student learning using exams and written reports. Disseminated knowledge of environmental laws, regulations and policies, and encouraged students to investigate current conservation efforts in the state of Texas.

Guest Lecturer, Huston-Tillotson University

Spring 2020

Terrestrial Field Biology (BIOL 3302), 20 students

Department of Biology

Invited to lead herpetological expedition into the Barton Creek Greenbelt, where students engaged in hands-on experience with native herpetofauna in Austin, TX, and learned the diversity of vertebrate anatomy and morphology.

Graduate Student Instructor, University of Michigan, Ann Arbor

2017 – 2019

Introduction to Biology Lab (BIO 173), 40 students

Department of Biology

Supported instructor by leading laboratory demonstrations to combine material from lecture and evaluated science comprehension through exams, written reports and presentations.

Pedagogical Training

Advanced Teaching Preparation Certificate, Faculty Innovation Center, UT Austin

Spring 2020

Participated in a seven-part series focused on developing core instructor skills that centered pedagogical- and cognitive learning theories, and inclusive teaching practices encourage student participation and learning

Supervised Teaching Course, Center for Research on Learning and Teaching, UM Ann Arbor

Fall 2017

Completed a semester long training series, which trainees were paired with senior graduate teaching assistants that assisted with developing strategies for effective lesson planning and provided in-class feedback on teaching

Mentoring Experience

Xavian Valadez-Rodela, Undergraduate Student, UT Austin

Summer 2024

Grant Maslowski, Volunteer, STRI

Summer 2022

(current: Conservation Biologist and Photographer, New Zealand Department of Conservation)

Eric de Framond-Bénard, Intern, STRI

Fall 2022

(current: Undergraduate Student, Université Rennes 1, France)

Nelson Guevara, Intern, STRI

Spring 2022

(current: President of the Biomundi Foundation)

Jorge Lopez, Intern, STRI

Fall 2021

(current: Undergraduate Student, Universidad de Panamá)

Dan Tatulescu, Undergraduate Student, UT Austin

Summer 2021

(current: Senior Undergraduate Student, UT Austin)

Malyne Wilkins, Undergraduate Student, UT Austin

2020 – 2021

(current: AmeriCorps Tutor & Mentor)

Aarcha Thadi, Intern, UM Ann Arbor

Summer 2018

(current: Graduate Student, University of Minnesota)

Research Experience

Graduate Research Fellow, Smithsonian Tropical Research Institute

2019 – 2024

Advisors: Drs. Rachel Page and Michael Ryan

- Collaborated and trained researchers at the Universidad de Panamá, Université Rennes 1, France, Universidad de los Andes, Mérida, Venezuela, The Ohio State University, Aarhus University, Denmark, McGill University, Max Planck Institute of Animal Behavior, University of Antwerp, Belgium to capture and study foraging behavior of bats
- Designed and managed multiple field experiments; analyzed and illustrated data trends using R Program and Procreate
- Captured and implanted wild bats with Passive Integrated Transponders and released bats for long-term populational research
- Collected morphology measurements, fur and DNA samples, and photography of neotropical bats; organized samples in the laboratory for long-term storage for collaborators
- Presented and communicated importance of research to potential donors, public audiences, and colleagues
- Participated in the development of the Strategic Plan in the Integrative Biology Department at UT Austin
- Engaged in public outreach events: “Bat Night” in Panama and “Science Under the Stars”, at UT Austin

Graduate Student Researcher, University of Michigan, Ann Arbor

2017 – 2019

Advisor: Dr. Alison Davis Rabosky

- Petitioned and collected data in support of the reintroduction of Michigan Arctic grayling, *Thymallus arcticus*, into the East and West branches of Maple River, in collaboration with the Department of Natural Resources and Little River Band of Ottawa Indians; presented results of evaluation to Michigan government officials at local symposium
- Collaborated with Georgia Southern University to collect cloacal samples of several herpetofauna for cross-ecosystem microbiome research
- Fixed herpetofauna specimens in neutral formalin solution and collected tissue samples from various herpetofauna species in Michigan, Peru, and Belize for UM Museum of Zoology Herpetological Expedition
- Conducted field research at the Los Amigos Biological Station in Peru and the Belize Foundation for Research & Environmental Education Research Institute
- Participated as a Frontiers Program Recruiter and Reviewer of applicants to the 2018 cohort

Undergraduate Research Assistant, Case Western Reserve University

2016 – 2017

Advisor: Dr. Michael Benard

- Examined the effects of water sewage construction activity on the migration patterns of small-mouthed salamander (*Ambystoma texanum*) populations in Ohio; presented results of evaluation at local symposium
- Captured salamanders using minnow traps and collected toe-clippings and pictures of underbellies for individual identification for long term capture-recapture research

Research Funding

Ecology, Evolution and Behavior Research Grant, UT Austin \$1500 award over 3 years	2020 – 2023
Graduate Research Fellowship, National Science Foundation \$34,000 award over 3 years	2020 – 2023
Dean's Strategic Scholars Fellowship, UT Austin \$31,000 award	2019 – 2020
Ecology and Evolutionary Biology Grant, UM \$2000 award	2018 – 2019
University of Michigan Biological Station Fellowship, UM \$1000 award	Summer 2018

Publications

Peer-Reviewed Research Articles

1. **Sealey, BA**, James, L, Cohen, G, Ryan, MJ, Page, RA. 2024. Rapid foraging risk assessments in the Panamanian Jamaican fruit eating bat, *Artibeus jamaicensis*. *Animal Behaviour*. Open access until Oct 11: https://authors.elsevier.com/a/1jeSu_4tl0d-N
2. **Sealey, BA**, Larson, JG, Moore, TY, Sánchez-Paredes, CM, Westeen, EP, Davis Rabosky, AR. 2024. Body size and predator cues structure variation in defensive displays of Neotropical calico snakes (*Oxyrhopus* spp.). *Ethology*. <https://onlinelibrary.wiley.com/doi/full/10.1111/eth.13439>
3. Davis Rabosky, AR, Moore, TY, Sánchez-Paredes, CM, Westeen, EP, Larson, JG, **Sealey, BA**, Bailey A. Balinski, BA. 2020. Convergence and divergence in anti-predator displays: A novel approach to quantitative behavioural comparison in Neotropical snakes. *Biological Journal of the Linnean Society*. <http://dx.doi.org/10.1093/biolinnean/blaa222>
4. Schell, C, Guy, C, Shelton, D, Campbell-Staton, S, **Sealey, BA**, Lee, D, Harris, N. 2020. Recreating Wakanda by promoting Black excellence in ecology and evolution. *Nature, Ecology and Evolution*. <https://doi.org/10.1038/s41559-020-1266-7>
5. **Sealey, BA**, Beasley, D, Halsey, S, Schell, CJ, Leggett, Z, Yitbarek, S, Harris, NC. 2020. Raising Black Excellence: Elevating Black ecologists through collaboration, celebration, and promotion. *Ecological Society of America*. <https://doi.org/10.1002/bes2.1765>
6. Crespo-Hernández, CE, Vogt, AR, & **Sealey, BA**. 2013. On the Primary Reaction Pathways in the Photochemistry of Nitro- Polycyclic Aromatic Hydrocarbons. *Modern Chemistry & Applications*, 1(3), 1–7. <http://dx.doi.org/10.4172/2329-6798.1000106>

Submitted or In-Preparation Research Articles

7. Holmes, IA, Martínez-Fonseca, JG, May, R, **Sealey, BA**, Cerda, PA, Grundler, MR, Westeen, EP, Nondorf, D, Larson, JG, Myers, CR, Hendry, TA. 2024. Increased host diversity limits bacterial generalism but may promote microbe-microbe interactions. *In Review*. <https://www.biorxiv.org/content/10.1101/2024.04.24.590977v2>
8. **Sealey, BA**, James, L, Ryan, MJ, Page, RA. The influence of hunger state in the foraging decisions of frugivorous bats. *In prep.*
9. **Sealey, BA**, James, L, Ryan, MJ, Page, RA. The effect of moonlight on fruit bat food preferences. *In prep.*

Presentations

Invited Talks

1. Evaluating antipredator behavior and food preferences in the Jamaican fruit-eating bat, *Artibeus jamaicensis*. Dissertation Defense. Austin, TX. 2024.
2. From here to there: How I got into animal behavior. Berea City Schools Alumni Scientist Speaker Series. Berea, OH. 2024. Virtual.
3. Moonlight avoidance behavior in *Artibeus jamaicensis*. Knörnschild Animal Behavior Lab. Free University Berlin, Germany. 2022. Virtual.
4. Foraging risk assessments in fringe-lipped bats (*Trachops cirrhosus*). STRI Behavior Seminar. Gamboa, Panama. 2021. Virtual.
5. Antipredator Behavior in Snakes. Career Day at Canyon Vista Middle School. Austin, TX. 2021 June 4. Virtual.
6. Predator cues drive variation in anti-predator displays of South American Calico snakes (*Oxyrhopus*). Brains, Behavior & Evolution Seminar. UT Austin, Austin, TX. 2019.
7. Dead or Alive: The Hognose Snake. Bioblitz in Gaylord County, MI. 2018.

Oral Presentations

8. Risk-assessments in the foraging behavior of *Artibeus jamaicensis*. Ecological Society of America Conference. Portland, OR. 2023.
9. Antipredator decision making in *Artibeus jamaicensis* using visual and auditory cues. Animal Behavior Society Conference. San Jose, Costa Rica. 2022.
10. Evaluating the East and West branches of the Maple River for the reintroduction of Artic Grayling (*Thymallus articus*). Frontiers Symposium. UM, Ann Arbor, MI. 2017.
11. Effects of Sediment from Recreational Horse Trails on Aquatic Macroinvertebrate Communities. REU Symposium. Samford University, Birmingham, AL. 2016.

Poster Presentations

12. Effects of anthropogenic pressure on small-mouthed salamander (*Ambystoma texanum*) populations. Research ShowCASE Senior Capstone Symposium. CWRU, Cleveland, OH. 2017.
13. Comparison of mating behaviors between species of *Callicebus donacophilus* and *Pithecia pithecia* at Cleveland Metroparks Zoo. Choose Ohio First: Research Poster Conference. Case Western Reserve University, Cleveland, OH. 2015.

Honors & Awards

Graduate Student Appreciation Award , The University of Texas at Austin	2024
Leadership Travel Grant , Ecological Society of America Conference (\$500)	2023
Travel Student Grant , Graduate Student Assembly (\$500)	2023
Diversity Travel Grant , Animal Behavior Society Conference (\$500)	2022
Dean's Honors List , Case Western Reserve University	2013 – 2017
Office of Multicultural Affairs Achievement of Academic Excellence , CWRU	2015 – 2016

Community Service

Scientist Volunteer , Barro Colorado Island (BCI) Alumni Visit, BCI, Panama	2023
Scientist Volunteer , Bat Show-n-Tell, OHR Visit, STRI, Gamboa, Panama	2022
Scientist Volunteer , Bat Night, Cambridge University Visit, STRI, Gamboa, Panama	2022
Scientist Volunteer , Bat Night, Public Outreach Event for Gamboa, STRI, Gamboa, Panama	2022
Scientist Volunteer , Phil Recchio visit, STRI Associate Director for Advancement & Outreach	2022

Scientist Volunteer , Lonnie Bunch III Visit, Secretary of the Smithsonian, Gamboa, Panama	2022
Tutor , America Reads, Estabrook Elementary, Ypsilanti, MI	2018 – 2019
Scientist Volunteer , BioBlitz, UMBS, Pellston, MI	2017
Tutor , Mary M. Bethune School, Cleveland, OH	2014 – 2017
Camp Counselor , TRIO Upward Bound, Cleveland, OH	2014
Camp Counselor , Camp Mi-Bro-Be, Berea OH	2013

Academic Service

Graduate Student Representative , Black Ecologist Section in ESA	2020 – 2023
Undergraduate Mentor , Fearless Leaders Institute, UT Austin	2019 – 2020
Science Educator , Science Under the Stars, Integrative Biology Department, UT Austin	2019 – 2020
Resident Assistant , Residence Life and Services, CWRU	2014 – 2017

Professional Experience

Animal Care , Lake Erie Nature and Science Center, Bay Village, OH	2016 – 2017
Intern , REU at Samford University, Birmingham, AL	Summer 2016
Animal Care , Cleveland Metroparks, Cleveland, OH	2015 – 2016
Intern , Engineering Education Summer Outreach Program, Cleveland, OH	Summer 2015